# An IoT-Enabled System for Real-Time Fuel Monitoring, GPS Tracking, and Anomaly Detection in Trucks Using Non-Parametric Unsupervised Machine Learning

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***Abstract*—The logistics and transportation industry continues to face challenges in fuel management, anomaly detection, and real-time fleet monitoring. Existing solutions often operate in silos, resulting in fragmented data, undetected fuel irregularities, and delayed maintenance responses. This study proposes an IoT-based, context-aware system designed to monitor real-time fuel levels, track GPS location, and detect anomalous fuel consumption in trucks using non-parametric unsupervised machine learning models, including Matrix Profile and Isolation Forest. The system leverages CAN Bus/OBD-II interfacing for ECU fuel data, GPS modules for location logging, LoRa and GSM modules for reliable IoT-based communication, and embedded alert hardware for driver rest and maintenance notifications. Sensor and vehicle data are synchronized and transmitted to a cloud-based dashboard where anomalies are automatically flagged, and actionable alerts are generated. The system will be evaluated in real-world driving scenarios to measure detection accuracy, false positive rates, and response latency. This integrated platform aims to enhance operational safety, improve fuel efficiency, and enable proactive, data-driven fleet management decisions.**

***Index Terms*—Fuel Monitoring, GPS Tracking, Anomaly Detection, Internet of Things, Unsupervised Machine Learning, Isolation Forest, Matrix Profile.**

## I. Introduction

THE transportation and logistics industry is a critical driver of economic development but is increasingly facing operational inefficiencies and security challenges, especially in fuel management and vehicle tracking. Fuel theft, inaccurate fuel monitoring, inefficient routing, and lack of real-time data remain persistent issues for fleet operators, particularly in developing countries. These challenges are exacerbated by the absence of scalable, integrated technological systems that provide reliable data-driven solutions [?], [?], [?].

Fuel is one of the most substantial and recurring expenses in logistics operations, with its accurate monitoring being essential to maintaining profitability and operational efficiency. Even minor inconsistencies in fuel tracking—such as errors in manual readings, unreported consumption, or unnoticed leakage—can accumulate into significant losses over time. Komal Shoukat Ali et al. (2018) [?] report that fuel-related fraud and inefficiencies can contribute up to 30% of the total operational overhead in large fleet operations. These losses are often compounded by issues such as unauthorized vehicle use, fuel siphoning, and unoptimized driving routes, which not only drive up fuel costs but also disrupt delivery schedules, vehicle maintenance cycles, and driver accountability. Despite efforts to manage these concerns, many logistics providers continue to rely on fragmented or outdated systems that offer limited visibility into fuel-related metrics and driver behavior [?], [?].

In addition to financial implications, traditional fuel monitoring systems struggle to detect abnormal fuel consumption patterns in real time. These anomalies may be caused by aggressive or inefficient driving habits, extended engine idling, mechanical malfunctions (such as injector faults or fuel line leaks), or variable environmental conditions like terrain and traffic. However, without intelligent tools that correlate fuel data with contextual factors—such as vehicle speed, route topography, and time of day—fleet managers are often unable to distinguish between acceptable fuel use and problematic deviations. As highlighted by Mumcuoglu et al. (2023) [?], conventional systems lack the sophistication to perform this level of analysis, leading to delayed detection of inefficiencies and reduced responsiveness. This underscores the urgent need for smart, data-driven solutions that combine fuel monitoring with GPS tracking and anomaly detection models to proactively manage consumption and enhance overall fleet performance [?], [?], [?].

To address these problems, researchers and engineers are increasingly exploring the convergence of IoT, GPS, and AI—particularly machine learning—to develop smart, real-time monitoring systems [?], [?], [?]. These systems aim to optimize fuel usage, enhance fleet security, and improve operational efficiency through automation and predictive analytics.

The core research gap, therefore, lies in the absence of a fully integrated, real-time, IoT-based system that combines fuel monitoring, GPS tracking, and anomaly detection powered by unsupervised machine learning. Addressing this gap is vital to improving operational efficiency, fuel security, and predictive maintenance in modern fleet management.

### *A. The Proposed Solution*

This study proposes the development of an integrated IoT-based system that unifies fuel level monitoring, GPS-based vehicle tracking, and unsupervised machine learning for anomaly detection. Unlike previous solutions that focus on isolated functions or static datasets, this system is designed with real-time operation in mind. During initial testing, data will be simulated using a custom-built fuel tank setup to enable controlled evaluation of the system's performance and anomaly detection capabilities.

For anomaly detection, the system will employ unsupervised, non-parametric machine learning algorithms like Matrix Profile and Isolation Forest. These models excel in identifying outliers within time-series data without the need for labeled datasets—making them ideal for fleet scenarios where data behavior varies with routes, driver habits, or environmental conditions. Unlike traditional parametric approaches, non-parametric models are more flexible and adaptive, as they do not assume any fixed data distribution.

In addition, a context-aware dashboard will be developed to provide fleet managers with meaningful insights by correlating fuel usage patterns with routes and travel times. The dashboard will support real-time notifications such as driver rest alerts, maintenance prompts, and efficiency summaries—enabling data-informed decision-making and operational improvements.

### *B. Research Objectives*

**General Objective:** To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines. The specific objectives are as follows:

**SO1:** To measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within $\pm 5\%$.

**SO2:** To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.

**SO3:** To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.

**SO4:** To design and evaluate a context-aware fuel usage deviation modeling approach using non-parametric unsupervised machine learning models, like Matrix Profile or Isolation Forest, achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.

**SO5:** To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.

## II. Related Work

The transportation and logistics sector has attracted significant scholarly attention, particularly in the development of smart systems for operational efficiency and security. The academic literature reflects a growing interest in leveraging IoT, GPS, and AI technologies to enhance fleet management, particularly in areas such as fuel monitoring, vehicle tracking, and anomaly detection. Several recent studies have investigated IoT-enabled fuel monitoring, GPS tracking, and anomaly detection solutions [?], [?], [?], [?].

**Intelligent IoT-Enabled Real-Time Monitoring System for Logistics Management:** Joshi et al. [?] developed a hybrid logistics management system that integrates IoT sensors, GPS, Radio Frequency Identification (RFID), and cloud analytics. Their solution achieved real-time tracking of assets and vehicles, enhancing transparency and decision-making in logistics operations. However, the study was limited by the type of hardware used and did not include advanced AI components for anomaly detection.

**Fuel Consumption Classification for Heavy-Duty Vehicles:** Mumcuoglu et al. [?] employed machine learning to classify fuel consumption anomalies based on naturalistic driving data from 57 trucks. With an accuracy of 92.2%, their approach effectively identified abnormal fuel consumption linked to driver behavior. Nevertheless, the dataset was relatively small (606 records), and integration with real-time vehicle diagnostics was not tested.

**Development of an Intelligent Fuel Monitoring and Theft Detection System:** Mathi et al. [?] introduced an Internet of Things (IoT)-based solution combining resistance-based fuel level sensors, GPS, Global System for Mobile Communications (GSM), and real-time alarm mechanisms. Their system successfully monitored fuel and battery anomalies in hybrid vehicles but lacked higher-precision sensors and deep learning integration for smarter detection.

Despite these advancements, several limitations and research gaps remain:

**Scalability Constraints:** Many existing solutions are limited to small-scale deployments or specific vehicle types (e.g., motorcycles, generators) and have not been validated for large truck fleets in diverse environments [?], [?].

**Data and Sensor Limitations:** Several studies rely on basic sensors (resistance, ultrasonic) that may lack the accuracy and robustness needed for high-volume commercial operations. Furthermore, datasets used are often small or simulated, limiting generalizability [?], [?].

**Fragmented Solutions:** Many systems only address a single issue—fuel monitoring, GPS tracking, or theft detection—without offering a comprehensive, unified solution integrating these components using AI and IoT [?], [?].

These gaps underline the need for a robust, scalable, and integrated IoT-enabled system capable of monitoring fuel levels in real time, tracking vehicle location, and detecting operational anomalies using machine learning algorithms. Such a system holds the potential to significantly reduce fuel fraud, enhance operational efficiency, and support more informed decision-making in fleet management.

## III. System Architecture and Conceptual Design

The system follows a layered architecture with a focus on modularity, scalability, and reliability. The conceptual framework integrates the following components:

**IoT Communication:** LoRa transceivers enable low-power, long-range data transmission, with cellular backup, balancing cost and bandwidth considerations.

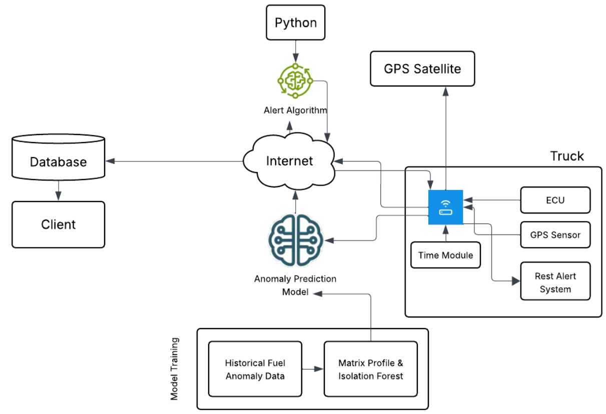


Fig. 1. Proposed conceptual diagram.

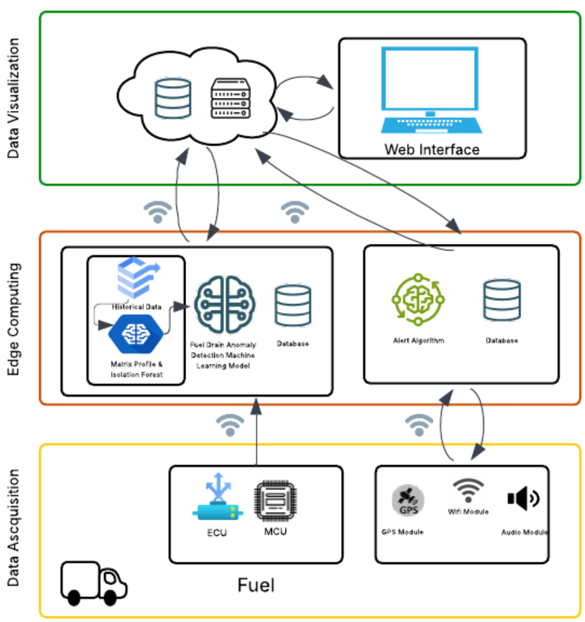


Fig. 2. Proposed system architecture diagram.

**Cloud Intelligence:** Historical data is used to train unsupervised ML models (Isolation Forest, Matrix Profile) to detect anomalies. The cloud evaluates incoming real-time data and triggers alerts.

**Client Visualization:** A web-based dashboard provides real-time visualization of fuel levels, routes, alerts, and historical playback for decision-making.

The system architecture follows a layered, end-to-end IoT design composed of three major layers: the data acquisition layer, the communication and cloud processing layer, and the data visualization layer. This structure supports real-time fuel monitoring, GPS-based route tracking, driver safety alerts, and cloud-based anomaly detection.

At the data acquisition layer, the vehicle is outfitted with digitally calibrated float sensors integrated through the truck's ECU/diagnostic connector to obtain accurate fuel-level measurements. The microcontroller (MCU) serves as the primary edge device, interfacing with the float sensor, GPS module (logging coordinates at 5-second intervals), rest alert subsystem, and time-tracking module. A LoRa transceiver functions as the main channel for transmitting fuel, location, and operating-hour data to the cloud, while the cellular module is used as fallback.

The communication and cloud processing layer handles real-time data ingestion and anomaly detection. Incoming packets from the vehicle are received through cloud-based API endpoints and routed to processing services where the trained machine learning model—developed offline using Matrix Profile or Isolation Forest—is deployed for fuel anomaly evaluation. A Python-based alert engine evaluates fuel–location–time patterns and stores synchronized measurements in a centralized database.

The data visualization layer consists of a web-based monitoring dashboard that retrieves information from the cloud database. Through this interface, clients can view real-time truck locations, fuel levels, distance traveled, driver operating hours, and alert notifications. The dashboard also supports historical playback of route and fuel usage trends, enabling improved operational awareness and proactive decision-making.

## IV. Methodology

The methodology integrates theoretical foundations on cyber-physical systems and IoT architectures with practical implementation strategies. The system leverages real-time sensor data, including digitally calibrated float sensors interfaced via the vehicle's CAN bus or OBD-II port, GPS modules for location tracking, and embedded alert subsystems for driver safety. These components feed into a microcontroller that preprocesses signals, synchronizes time-series data, and transmits information to a cloud platform primarily via LoRa, with GSM as a backup channel.

Offline, historical and hybrid datasets, combining fleet telemetry and publicly available vehicle logs, are used to train non-parametric unsupervised machine learning models such as Matrix Profile and Isolation Forest. These models identify anomalous fuel consumption patterns and provide context-aware alerts. In the cloud, a Python-based engine continuously evaluates incoming data, logs anomaly events, and stores synchronized measurements in a centralized database. Users access the information through a web-based dashboard that displays real-time truck status, historical fuel usage, route playback, and operational alerts.

The methodology includes signal preprocessing (smoothing via moving averages), time synchronization using linear interpolation, hybrid dataset construction, and rigorous evaluation metrics (precision, false positive rate, accuracy, detection latency) to validate system objectives. The approach follows iterative, agile development cycles, allowing continuous refinement of sensor integration, communication protocols, and anomaly detection algorithms, thereby ensuring reliability, scalability, and actionable fleet monitoring insights.

### A. Data Acquisition and Preprocessing

Data acquisition and preprocessing are crucial for reliable anomaly detection. The methodology includes the following steps:

**Sensor Calibration:** Fuel-level sensors are calibrated using known volumes, verified with manual measurements, and cross-checked with historical data trends.

**Signal Filtering:** Raw sensor signals are filtered using moving-average and median filters to reduce noise caused by vehicle motion or fuel sloshing.

**Synchronization:** Fuel, GPS, and operating-hour data are timestamped and synchronized to ensure consistent data alignment for model evaluation.

**Feature Engineering:** Derived features include fuel consumption rate, distance-to-fuel ratio, route segment performance, and driver behavior metrics, which feed into anomaly detection models.

Sensor signals can exhibit noise, quantization error, and environmental fluctuation. To improve stability, the system uses a Simple Moving Average (SMA) filter:

$$\mathrm{SMA}_t = \frac{1}{n}\sum_{i=0}^{n-1} x_{t-i}. \tag{1}$$

This reduces high-frequency noise and creates a clearer trend for anomaly detection algorithms. Sensor streams often arrive at irregular timestamps, and synchronizing them requires interpolation theory. For two samples at timestamps $T_i$ and $T_{i+1}$, the interpolated point at $t_j$ is computed as:

$$x(t_j) = x(T_i) + \frac{x(T_{i+1}) - x(T_i)}{T_{i+1} - T_i}(t_j - T_i). \tag{2}$$

This ensures aligned and consistent time-series data across fuel level, GPS speed, and distance.

### B. Anomaly Detection and Machine Learning

The anomaly detection workflow is as follows:

**Offline Model Training:** Historical datasets are used to train Isolation Forest and Matrix Profile models to detect abnormal fuel usage patterns.

**Context-Aware Thresholding:** Thresholds are dynamically adjusted based on vehicle type, route segment, and driver behavior to reduce false positives.

**Evaluation Metrics:** Precision, FPR, accuracy, and detection latency are calculated; confusion matrices, ROC, and PR curves visualize performance.

## V. Evaluation

To evaluate these models, the outputs are validated against manually labeled ground truth data. From this, the number of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) are counted. Key metrics are computed as follows.

**Precision**

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{3}$$

Precision reflects how many detected anomalies are truly anomalous. A high precision means fewer false alarms, which is important for maintaining trust in the alerting system.

**False Positive Rate (FPR)**

$$\text{FPR} = \frac{FP}{FP + TN}. \tag{4}$$

FPR quantifies the likelihood of incorrectly classifying normal behavior as anomalous. Lower FPR values are desired to reduce noise in the system's alerts.

**Accuracy**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{5}$$

Accuracy provides an overall view of model correctness but can be misleading in imbalanced datasets, which is why it is used alongside precision and FPR.

**Detection Latency**

$$L_{\text{det}} = T_{\text{detected}} - T_{\text{actual}}. \tag{6}$$

Detection latency measures the difference between the time an anomaly is detected and the time it actually occurred. The objective is achieved if the unsupervised system attains at least 90% precision (indicating that the majority of flagged events are correct), a false positive rate not exceeding 10%, and average detection latency under 10 seconds. For interpretability, confusion matrices and performance plots such as ROC and PR curves will be used. Detection events will also be overlaid on synchronized GPS location maps and fuel-level graphs for context-aware validation.

## VI. Conclusion

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